

Fathoms — RTS Design Document

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Overview

Fathoms is a 2–6 player steampunk island RTS in Depth's Deep Arcade, built for 20–30 minute matches over LAN or the internet. Players start on a home island, claim new islands with Lighthouses, fight over Coal, Ichor, and relics, and bargain with or battle pirates, tribes, and volcano gods.

Design pillars

- **Islands are the map.** Every expansion and every attack crosses water, so the navy decides where you can go and the army decides what you keep.
- **Six factions that play differently.** Each has a signature mechanic that changes how you build, fight, or expand — not just a stat bonus.
- **Neutral powers make every match a story.** Pirates can be hired against rivals, tribes raid or ally, volcanoes reward the brave and punish the careless.
- **Short and readable.** Five resources, three eras, about 10 shared unit types per faction, and a hard 30-minute cap.
- **Rooted in Depth.** Factions, bosses, and relics come from the roguelike's locations and characters, so the arcade feels like part of the same world.

Reference games and what Fathoms takes from each

Game	What Fathoms borrows
Age of Empires II	Workers, three eras with building requirements, unit counter triangles, relics, unique units per faction, idle-worker hotkeys, market trading
Stellaris	Claiming territory with outposts, chokepoint defense, pick-one-of-three tech cards, diplomatic stances
Civilization	Founding colonies on new land, tile improvements around settlements, city-state style neutral towns

Game	What Fathoms borrows
Call of War	Terrain modifiers, morale and supply away from home, stack efficiency, upkeep, diplomacy stances, AI takeover of inactive players
Polytopia	Short matches, compact tech, score victory at the time limit, small readable maps
Pharaoh (Ozymandias)	Housing caps and base needs, gods that bless or punish
Depth (roguelike)	Factions, bosses, relics, and status effects (stun, bleed, poison, weaken, strengthen)

Match setup and lobby

The host picks every match setting in the lobby; each player picks a faction, a color, and a team, then pulls their ready lever.

Setting	Options	Default
Players	2–6 humans, empty slots filled by AI or left closed	2
Teams	Free-for-all, 2v2, 3v3, 2v2v2, 4v2 (handicapped), or locked custom teams	Free-for-all
Map size	Small, Medium, Large (auto-picked from player count, host can override)	Auto
Map type	Archipelago, Twin Continents (two big landmasses plus small islands), Ring of Fire (volcano center), Scattered Isles (many tiny islands)	Archipelago
Victory	Conquest, Relics, Volcano Ascension, Score only; any combination can be on	Conquest + Relics + Volcano, Score at time cap
Time cap	20, 25, 30, 40 minutes, or none	30

Setting	Options	Default
Starting resources	Standard, High, Very High	Standard
Starting era	Sail, Steam	Sail
Neutral powers	Pirates, Tribes, Volcanoes, Kraken each on or off; strength Low / Normal / High	All on, Normal
Peace timer	Off, 3, 5, 8 minutes (no player-vs-player attacks allowed)	Off
Diplomacy	Locked teams or Open diplomacy (players can change stances)	Locked in team games, Open in FFA
Reveal map	Normal fog, Explored, All visible	Normal fog
Game speed	0.75×, 1×, 1.25×, 1.5×	1×
AI difficulty	Deckhand (easy), Mate, Captain, Admiral (hard)	Mate
Random seed	Random or typed in (lets friends replay a map they liked)	Random

Faction selection rules. Duplicate factions are allowed. A Random pick shows a question mark until the match starts. Each player sees the others' picks only when the host locks the lobby, so no one counter-picks at the last second.

Handicaps. The host can give any player +10% to +30% gather rate. This keeps games fun when skill levels differ a lot among friends.

Map and archipelago generation

Maps are tile grids generated from a seed with rotational symmetry, so every player gets the same home island and the same distances to key sites. The center of the map holds the richest and most dangerous sites.

Map sizes

Size	Players	Grid (tiles)	Islands	Volcanoes	Pirate coves	Tribal towns	Sunken ruins (relics)	Coal islands
Small	2	128 × 128	12	1	1	2	3	4
Medium	3-4	176 × 176	20	1	2	4	5	6-8
Large	5-6	224 × 224	30	2	3	6	7	10-12

One tile is about the width of a worker. A worker crosses a Medium map's full width in about 3 minutes on foot-equivalent speed; a Steam ship does it in about 75 seconds.

Island types

Island	Size (tiles across)	What it holds	Who wants it
Home island	30–36	Harbor, 8 fertile tiles, 2 Brass seams, 1 small Coal seam, 1 kelp bed offshore, 6 fish schools	Its owner; losing it is usually losing the game
Fertile isle	14–20	Farmland, fruit groves, fish	Food-hungry players, Crustaceans
Mining isle	12–18	2–3 Brass seams, rocky hills	Everyone mid-game
Coal isle	10–14	2 Coal seams	Clockwork, anyone with a navy
Reef atoll	6–10 (mostly shallows)	Kelp beds, pearl banks (Doubloons)	Merfolk, fishing economies
Ruin isle	8–12	One relic in a sunken temple, guarded	Atlanteans, Relic victory
Tribal isle	16–22	A tribal town, jungle, fertile land	Islanders, anyone wanting loot
Pirate cove	12–16	A pirate fortress and black market	Everyone, for deals
Volcano isle	20–26	Ichor vents, obsidian, the summit altar, the Sun God	Late-game players, Volcano victory

Terrain and its effects

Terrain	Movement	Combat effect	Notes
Deep sea	Ships 100%, land units cannot enter	—	Merfolk units can swim here
Shallows	Ships 70%, land units wade at 50%	Wading land units take +20% damage	Crustaceans walk here at full speed; ironclads cannot enter

Terrain	Movement	Combat effect	Notes
Beach	Land 100%	Units that landed in the last 10 s take +25% damage	Where amphibious assaults happen
Grassland	Land 100%	—	Best building ground
Jungle	Land 70%	Ranged attacks into jungle deal -25%; units inside are hidden until an enemy is within 4 tiles	Ambush terrain
Hills and cliffs	Land 60%	Defenders on higher ground deal +25% and take -15%	Towers here get +2 range
Mountains	Impassable	—	Blocks line of sight
Lava flow	Impassable while active	40 damage per second	Only during eruptions
Kelp forest (sea)	Ships 80%	Hidden like jungle	Spreads if Merfolk own it
Currents	One-way, +50% ship speed	—	Fixed lanes that connect regions

Weather rolls every 3 minutes on Medium and Large maps: calm (60%), fog bank (20%, vision -50% in its area), storm (15%, ships in it take 2 damage per second and move at 60%), whirlpool (5%, pulls ships toward its center for 45 s). A storm or whirlpool never spawns within 20 tiles of a home island.

Fairness rules the generator must follow

1. Home islands sit on a circle around the center, evenly spaced by angle.
2. Each player's nearest Coal isle, Mining isle, and Fertile isle are the same distance (± 2 tiles) from their Harbor.
3. Each player has one tribal town and one pirate cove at the same distance, or shares an equidistant one with a neighbor.
4. Volcanoes and the richest ruins sit in the center, equally far from everyone.

5. The generator rejects and rerolls any map that fails a distance check; the seed is shown in the lobby.

Resources and economy

Fathoms uses four gathered resources plus Doubloons, a trade-only currency for pirates and mercenaries. Workers gather by walking to a resource and carrying it back to a drop-off building, as in AoE2, so drop-off placement matters.

Resource	Main uses	Sources	Base gather rate (per worker or boat per second)	Carry load
Food	Workers, infantry, riders, era advance	Fish schools (fishing boat), farms, fruit groves, kelp beds (boat)	Fish 0.50 · Farm 0.40 · Grove 0.45 · Kelp 0.35	10
Brass	Buildings, ships, ranged and heavy units	Brass seams, wrecks (salvage)	Seam 0.40 · Wreck 0.50	10
Coal	Steam ships, airships, machines, Clockwork units, some techs	Coal seams	0.35	8
Ichor	Leviathan Era advance, elite units, ultimates, top techs	Ichor vents (ruin isles, volcanoes), relics (trickle)	Vent 0.15 · Relic 0.30 per relic, no worker needed	5
Doubloons	Pirate deals, mercenaries, ransoms, bribes	Pearl banks (fishing boat), trade routes, salvage bonus, tribal loot, raiding enemy trade ships, selling at the Exchange	Pearls 0.20	5

Resource amounts per node

Node	Amount	Notes
Fish school	450	Regrows 1 fish per 5 s up to its cap if at least 1 fish is left
Farm	Unlimited	Costs 60 Brass; 1 worker per farm
Fruit grove	400	Does not regrow
Kelp bed	800	Regrows 1 per 4 s; Merfolk gather it at 0.50
Brass seam	2,500	Home islands have 2
Wreck	600 Brass + 30 Doubloons	Some wrecks are trapped (see neutral powers)
Coal seam	2,000 (home seam 800)	
Ichor vent	800 (ruin isle), 3,000 (volcano)	Volcano vents sit in eruption paths
Pearl bank	300 Doubloons	Regrows 1 per 10 s

Starting position (Standard). Harbor, 4 workers, 1 fishing boat, 1 scout skiff, 200 Food, 200 Brass, 50 Coal, 0 Ichor, 50 Doubloons. High doubles these; Very High gives 1,000 of each gathered resource and 200 Doubloons.

Population. The Harbor supports 10 population. Each Cottage adds 5 (cost 30 Brass). The cap is 100 in 2-player games, 80 at 3–4 players, and 60 at 5–6 players, so six-player games stay readable and network-light. Workers and infantry take 1 population, riders and heavy units 2, ships 2–4, airships 4.

Upkeep. Steam-powered units burn Coal every minute (listed in each unit's stats). If your Coal stockpile hits 0, all Coal-burning units move 40% slower and deal 25% less damage until it recovers. Sail-era ships and ordinary soldiers cost no upkeep. This is Call of War's oil rule, and it makes Coal isles worth fighting for.

The Exchange. Trade Food, Brass, and Coal for each other. Every trade starts at 100 for 70 and moves that resource's price 3% up (buying) or down (selling); prices drift back 1% every 20 s. Ichor cannot be bought here, only from pirates. Any resource can be sold for Doubloons at 100 for 15.

Economic targets for a steady player (used to tune costs and to build the AI): about 20 workers at minute 6, 32 at minute 12, 45 at minute 18. At 32 workers plus 4 fishing boats, income is roughly 300 Food, 170 Brass, and 90 Coal per minute after walking time (about 70% of the base rate).

Buildings

Every faction shares these buildings; factions rename and reskin them and swap in their own where noted in the faction sections. Workers build them; build time halves with each extra worker, down to a minimum of one third.

Building	Era	Cost	HP	Build time	Role
Harbor	Start (one free)	350 Brass, 100 Coal	2,400	150 s	Town center: trains workers, fishing boats, scout skiffs; drop-off for everything; +10 pop; garrisons 15 workers and fires 1 shot per garrisoned unit; can train Transports
Cottage	Sail	30 Brass	400	20 s	+5 population
Farmstead	Sail	80 Brass	800	30 s	Food drop-off; farm and fishing techs
Mine Shed	Sail	80 Brass	800	30 s	Brass, Coal, and Ichor drop-off; mining techs
Farm	Sail	60 Brass	300	15 s	Infinite Food, 1 worker
Dock	Sail	120 Brass	1,500	40 s	Fishing boats, transports, warships, naval techs; Food and Brass drop-off
Barracks	Sail	150 Brass	1,500	45 s	Riflemen, Harpooners; later Bell Guards, Medics
Stable	Sail	150 Brass	1,500	45 s	Scout Riders; Lancers in Steam
Lighthouse	Sail	120 Brass, 20 Coal	1,200	60 s	Claims its island (radius 14 tiles), vision 16, supply zone for morale; only one per island; the first one on a new island needs a worker delivered by boat
Watchtower	Sail	60 Brass, 30 Food	700	40 s	Range 7, attack 6; garrisons 4

Building	Era	Cost	HP	Build time	Role
Palisade	Sail	2 Brass per tile	150 per tile	3 s per tile	Cheap wall
Stone Seawall	Steam	6 Brass per tile	1,200 per tile	8 s per tile	Heavy wall; blocks landings on beaches
Colony Hall	Steam	250 Brass, 50 Coal (plus Colony Ship)	1,800	100 s	Second town center on a claimed island: trains workers, drop-off for everything, +8 pop; max 3 per player
Workshop	Steam	175 Brass	1,500	50 s	Combat upgrades (Forge line)
Siege Works	Steam	200 Brass, 50 Coal	1,500	60 s	Mortars
Exchange	Steam	175 Brass	1,500	50 s	Resource trading, Trade Ships, Doubloon sales
Coastal Battery	Steam	150 Brass, 60 Coal	1,800	60 s	Range 9 gun that deals +100% to ships; garrisons 5
Coaling Station	Steam	120 Brass	1,000	40 s	Extends coaling range 30 tiles; Coal drop-off from boats
Chapel of the Deep	Steam	150 Brass, 50 Ichor	1,200	60 s	Relic storage (relics only count when garrisoned here or at a Harbor); faction healer units
Airship Hangar	Leviathan	250 Brass, 150 Coal	1,800	80 s	Airships
Flak Tower	Leviathan	120 Brass, 60 Coal	1,000	50 s	Range 10 anti-air, +300% to airships; cannot hit ground
Faction Wonder	Leviathan	1,000 Brass, 500 Coal, 300 Ichor	6,000	240 s	Each faction's great work; grants its passive and counts as 150 score (see factions)

Garrisoning. Workers, infantry, and Medics can hide in Harbors, Colony Halls, Watchtowers, and Coastal Batteries. Garrisoned units heal 1 HP per second and add firepower.

Losing a Lighthouse. The island stays yours until an enemy builds their own Lighthouse there; your buildings on it keep working but units there lose morale (see combat rules).

Eras and technology

There are three eras. Advancing is researched at the Harbor, needs two buildings from the current era, and pauses worker training at that Harbor while it runs.

Advance	Cost	Time	Requires two of	Target timing
Sail → Steam	400 Food, 150 Brass	90 s	Barracks, Stable, Dock, Farmstead, Mine Shed, Lighthouse on a second island	Minute 7–9
Steam → Leviathan	800 Food, 600 Brass, 150 Ichor	120 s	Workshop, Siege Works, Exchange, Coastal Battery, Colony Hall, Chapel of the Deep	Minute 16–19

The first Ichor comes from your nearest ruin isle's vent (800 Ichor, equidistant for every player), so the Leviathan race always starts with a fight or a deal over ruins.

Economy technologies

Tech	Building	Era	Cost	Effect
Cast Nets	Farmstead	Sail	100 Food, 50 Brass	Fishing +15%
Iron Plows	Farmstead	Sail	125 Food, 75 Brass	Farms +15%
Pickaxes	Mine Shed	Sail	100 Food, 75 Brass	Brass and Coal mining +15%
Handcarts	Harbor	Sail	150 Food, 50 Brass	Workers carry +3 and move +10%
Trawlers	Farmstead	Steam	200 Food, 150 Brass	Fishing +15%; fishing boats +20% HP
Steam Threshers	Farmstead	Steam	250 Food, 150 Brass, 50 Coal	Farms +15%
Pneumatic Drills	Mine Shed	Steam	200 Food, 150 Brass, 50 Coal	Mining +15%
Ichor Condensers	Mine Shed	Steam	200 Brass, 100 Coal	Ichor gathering +25%
Coal Efficiency	Dock	Steam	150 Brass, 100 Coal	Coal upkeep –33%

Tech	Building	Era	Cost	Effect
Guild Charters	Exchange	Steam	200 Food, 200 Brass	Exchange price penalty halved; Trade Ship income +20%
Letters of Marque	Exchange	Steam	150 Brass, 60 Doubloons	Pirate hiring -20%; your raiders capture +50% loot
Clockwork Harvesters	Farmstead	Leviathan	300 Food, 200 Brass, 100 Coal	Farms +20%
Deep-Bore Rigs	Mine Shed	Leviathan	300 Food, 200 Brass, 100 Coal	Mining +20%; seams last +25%

Military technologies (the Forge line at the Workshop). Each line has three levels: level 1 in Steam, levels 2-3 in Leviathan. Each level costs 150 / 300 / 450 Food and 100 / 225 / 350 Brass.

Line	Per level
Cutlasses	Melee units +1 attack
Rifling	Ranged land units +1 attack, +1 range at level 3
Riveted Plate	Land units +1 melee armor
Oilskin Coats	Land units +1 pierce armor
Naval Gunnery (Dock)	Ships +1 attack, +1 range at level 3
Armored Hulls (Dock)	Ships +10% HP, +1 pierce armor

Other technologies

Tech	Building	Era	Cost	Effect
Fresnel Lens	Lighthouse	Sail	75 Brass, 25 Coal	Lighthouse vision +6, claim radius +3
Supply Caches	Lighthouse	Steam	150 Food, 100 Brass	Morale loss outside territory halved

Tech	Building	Era	Cost	Effect
Field Medicine	Barracks	Steam	200 Food, 100 Brass	Medics heal +50%; infantry regen 0.5 HP/s out of combat
Copper Sheathing	Dock	Steam	200 Brass, 50 Coal	Ships +10% speed
Fortified Masonry	Workshop	Steam	200 Brass	Buildings +20% HP; towers +1 range
Amphibious Drills	Barracks	Steam	150 Food, 150 Brass	Removes the landing penalty; transports unload 2× faster
Elite upgrades	Unit's building	Leviathan	About 2× the unit's cost + 25 Ichor	Each unit line becomes Elite: +20% HP, +15% attack
Faction techs	Varies	Steam and Leviathan	See factions	Two per faction

Every tech costs the same for every faction, so no faction can out-tech another by default. Faction differences live in their mechanics and unique content.

Shared units and combat rules

Every faction has this roster (reskinned to fit it) plus its own unique units. Stats below were tuned in a combat simulator so each counter wins an equal-cost fight with about 55–65% of its army left (see Balance).

Damage formula. Damage = attack + bonus against the target's tags – the target's armor of the matching type, minimum 1. There are three damage types: melee (reduced by melee armor), pierce (reduced by pierce armor), and blast (ignores armor, and deals 50% splash within 1.5 tiles). Speeds are in tiles per second; ranges in tiles.

Land units

Unit	Era	Building	Cost	Pop	HP	Attack	Reload	Range	Speed	Armor (melee/pierce)	Tags	Bonuses	Train
Worker	Sail	Harbor	50 F	1	25	3 melee	2.0	0.5	1.0	0/0	Infantry	—	20 s

Unit	Era	Buildin g	Cost	Pop	HP	Attack	Reload	Range	Speed	Armor (melee/ pierce)	Tags	Bonuse s	Train
Riflema n	Sail	Barrack s	40 F, 30 B	1	50	7 pierce	2.0	5	1.0	0/1	infantr y, ranged	+1 vs infantr y	22 s
Harpoo ner	Sail	Barrack s	35 F, 25 B	1	55	4 melee	2.0	0.5	1.05	1/3	infantr y	+8 vs large	20 s
Scout Rider	Sail	Stable	80 F	1	70	4 melee	2.0	0.5	1.8	0/2	large, moun ted	vision 10	25 s
Lancer	Steam	Stable	60 F, 45 B	2	120	9 melee	1.8	0.5	1.6	2/2	large, moun ted	+4 vs ranged, +15 vs siege	30 s
Bell Guard	Steam	Barrack s	50 F, 65 B	2	150	10 melee	2.0	0.5	0.8	3/5	infantr y, large	+6 vs buildin gs	32 s
Medic	Steam	Barrack s	40 F, 40 B	1	40	Heals 2 HP/s	—	4	1.0	0/1	infantr y	Can cure bleed and poison	25 s
Mortar	Steam	Siege Works	100 B, 50 C	3	80	14 blast	5.0	10 (min 3)	0.6	0/4	siege, large	+45 vs buildin gs; upkeep 2 Coal/mi n	40 s

Naval units

Unit	Era	Buildin g	Cost	Pop	HP	Attack	Reload	Range	Speed	Armor (melee/ pierce)	Tags	Bonuse s	Upkeep (Coal/ min)
Fishing Boat	Sail	Harbor, Dock	60 B	1	60	—	—	—	1.8	0/0	ship, light	Gather s fish, kelp, pearls	0
Scout Skiff	Sail	Harbor, Dock	50 B	1	80	—	—	—	3.0	0/1	ship, light	Vision 12	0
Transp ort	Sail	Harbor, Dock	100 B	1	200	—	—	—	1.6	0/3	ship, light	Carries 10 pop	0
Sloop	Sail	Dock	90 B	2	130	7 pierce	2.5	6	1.8	0/1	ship, light	Can refit into a Gunbo at for the cost differe nce	0

Unit	Era	Building	Cost	Pop	HP	Attack	Reload	Range	Speed	Armor (melee/pierce)	Tags	Bonuses	Upkeep (Coal/min)
Colony Ship	Steam	Dock	150 F, 150 B	3	250	—	—	—	1.4	0/2	ship	Carries 3 workers; unpacks into a Colony Hall site	0
Trade Ship	Steam	Exchange	100 B	2	180	—	—	—	1.8	0/2	ship, light	See Diplomacy	0
Gunboat	Steam	Dock	110 B, 25 C	2	220	10 pierce	2.2	7	2.0	0/2	ship, light	+3 vs torpedo, +8 vs buildings	2
Torpedo Boat	Steam	Dock	100 B, 40 C	2	160	26 blast	5.0	5	2.4	0/1	ship, light, torpedo	+26 vs heavy	2
Ironclad	Steam	Dock	200 B, 80 C	4	470	18 pierce	3.0	7	1.3	0/6	ship, heavy	+3 vs light; can bombard land 7 tiles inland	5

Air units

Unit	Era	Building	Cost	Pop	HP	Attack	Reload	Range	Speed	Notes	Upkeep (Coal/min)
Airship	Leviathan	Airship Hangar	180 B, 100 C, 20 I	4	260	30 blast	4.0	1 (drops bombs below it)	1.5	Flies over everything; cannot hit other airships; only Riflemen, Gunboats, Coastal Batteries, and Flak Towers can hit it (Riflemen and Gunboats at 50% damage)	6

Counter triangles

Unit	Strong against	Weak against
Rifleman	Harpooner	Lancer, Bell Guard
Harpooner	Lancer, Bell Guard, Scout Rider	Rifleman
Lancer	Rifleman, Mortar	Harpooner, Bell Guard
Bell Guard	Rifleman, Lancer, buildings	Harpooner, Mortar
Mortar	Buildings, clumped infantry	Lancer, anything that reaches it
Gunboat	Torpedo Boat, buildings, shore	Ironclad
Torpedo Boat	Ironclad	Gunboat, Sloop
Ironclad	Gunboat, shore targets	Torpedo Boat
Airship	Buildings, slow ground units	Flak Tower, Coastal Battery

Morale and supply (Call of War's rule)

- Every unit has morale from 0 to 100. Inside your territory (any Lighthouse, Harbor, or Colony Hall radius) it recovers 5 per 10 s.
- Outside your territory it drops 2 per 10 s; enemy territory doubles that.
- Below 50 morale: -10% damage. Below 25: -25% damage and -10% speed.
- Medics, heroes, and the Supply Caches tech slow the loss. An army can push about 4 minutes into enemy land before it weakens.
- Ships beyond coaling range (30 tiles from a friendly Harbor, Dock, or Coaling Station) move 25% slower.

Landings. A Transport unloads 1 unit per second on a beach. Units that landed in the last 10 s take +25% damage (Amphibious Drills removes this). Landing directly under a Watchtower or Coastal Battery is meant to hurt.

Status effects (from Depth's roguelike, used by unique units and heroes): Stun (cannot act), Bleed and Poison (damage over time; Medics cure them), Weaken (-20% attack or speed), Strengthen (+20% attack or speed). Durations are 3-8 s; the same effect does not stack, it refreshes.

Control. Box select, right-click to act, Ctrl+number groups, formations (line, box, staggered), stances (aggressive, defensive, stand ground, passive), patrol, attack-move, idle-worker and idle-army hotkeys, and waypoints for trained units, following AoE2's

hotkey layout.

The six factions

Each faction has one signature mechanic, two unique units, one hero, two faction techs, one ultimate, and a wonder. They are balanced to equal overall strength but peak at different times, so each has a window where it should attack and a window where it should defend.

At a glance

Faction	Playstyle	Signature mechanic	Strongest era	Main weakness
Nautilus Crew	Flexible, technological	Nemo's Logbook: pick 1 of 3 tech cards each era	Leviathan	Every unit costs 10% more
Islanders	Raiders and diplomats	Kinship: turn tribal towns into allies	Sail and Steam	Weak navy late
Crustacean Brood	Slow, tough, snowballing	Molting and seafloor walking	Steam and Leviathan	Slow, limited navy
Merfolk of the Weeds	Sea control and ambush	Kelp Spread and swimming units	Sail and Steam	Weak far inland
Atlantean Lost Ones	Slow start, huge finish	Devotion and conversion	Leviathan	Slow early economy
Clockwork Foundry	Industrial grind	Coal-powered army that never breaks	Steam	Collapses without Coal

Rules every faction shares. A hero costs 200 Food, 200 Brass, 50 Coal, is trained at the Harbor in the Steam Era, has 500 HP, and only one can exist; if it dies it can be revived at the Harbor after 60 s for half its cost. An ultimate is summoned in the Leviathan Era for 300 Ichor, stays for 120 s, has 2,500 HP and roughly 40 damage per second, and has a 6-minute cooldown after it leaves. Unique units follow the same damage rules as shared units.

Nautilus Crew

Captain Nemo's crew: a steampunk navy that can adapt to any plan. The easiest faction to learn.

- **Nemo's Logbook.** On reaching each era (including the start), draw 3 cards from the Logbook and keep 1. Cards include Galvanic Batteries (ships +10% damage), Deep Survey (reveal every ruin and wreck), Pressure Hulls (Nautilus Submarines +25% HP), Diving Bells (Ichor gathering +30%), Steam Turbines (ships +15% speed), Marine Drill (Diving Marines +2 armor), Surveyor's Instinct (Lighthouses cost 30% less), and Quartermaster (Coal upkeep -25%). No card may be drawn twice.
- **Weakness.** All units cost 10% more.

Unique unit	Era	Cost (before +10%)	Pop	HP	Attack	Range	Speed	Special
Nautilus Submarine	Steam	180 B, 70 C	3	260	30 blast, +20 vs heavy, reload 5.5	5	2.0 (1.4 submerged)	Dive: invisible while submerged; surfaces for 3 s after firing. Seen by Gunboats, Scout Skiffs, and Coastal Batteries within 4 tiles. Upkeep 3 Coal/min
Diving Marine	Steam	50 F, 40 B	1	75	8 pierce, reload 2.0	4	1.0	Amphibious: swims up to 10 tiles of water; no landing penalty

- **Hero: Captain Nemo.** Aura: nearby units +10% attack. Active "Full Ahead" (60 s cooldown): ships within 8 tiles +40% speed for 10 s.
- **Faction techs.** Ballast Tanks (Steam, 200 B, 100 C): Submarines can fire once every 20 s without surfacing. Electric Harpoons (Leviathan, 250 F, 200 B, 50 I): Harpooners stun targets for 1 s on hit, once every 8 s.
- **Ultimate: The Nautilus.** The legendary submarine surfaces for 120 s: 2,500 HP, torpedo volleys that deal splash damage, can submerge for 10 s once.
- **Wonder: Nautilus Drydock.** Your ships regenerate 2 HP/s inside your territory.

Islanders

The Dethroned Island Queen's people. They win by making the map's tribes fight for them and raiding with fast canoes.

- **Kinship.** A Shaman (Barracks, 60 F, 40 B) can perform a 45 s ritual at a tribal town with a gift of 150 Food and 100 Doubloons. The town becomes an ally: its island counts as Islander territory and it produces 1 free Tribal Warrior (60 HP, 7 melee attack) every 60 s, up to 6 per town. Warriors don't use population. If the Shaman is killed during the ritual, the gift is lost.
- **Totems** (80 Brass, one per island): friendly units within 8 tiles get +15% attack.
- **Economy bonus.** Fishing and fruit gathering +8%.
- **Weakness.** Ironclads have 15% less HP.

Unique unit	Era	Cost	Pop	HP	Attack	Range	Speed	Special
War Canoe	Sail	50 F, 45 B	1	110	6 pierce, reload 2.2	4	2.4	Carries 5 infantry and fights; no upkeep
Dog Handler	Sail	45 F, 25 B	1	65	6 melee, reload 1.5, +3 vs ranged	0.5	1.3	Attacks cause Bleed (2 dmg/s for 4 s)

- **Hero: The Coconut Queen.** Aura: nearby units heal 1 HP/s. Active "Coconut Barrage" (45 s cooldown): 40 blast damage in a 3-tile circle, stuns for 2 s.
- **Faction techs.** Blood Oath (Steam, 200 F, 150 B): Tribal Warriors +25% HP. Sun Drums (Leviathan, 250 F, 200 B): Totem radius +4, Totems also give +1 armor.
- **Ultimate: The Tribal Demigod.** A giant warrior that leaps between targets and knocks units back.
- **Wonder: Temple of the Sun.** All allied tribal towns produce warriors twice as fast; volcano Sun Gods ignore your units.

Crustacean Brood

The undersea cave's creatures: slow and armored, getting stronger with every fight they survive.

- **Molting.** Units earn experience equal to damage dealt. At 60 experience a unit molts: it is cocooned for 5 s (takes +50% damage), then gains +25% HP and +15% attack and a bigger model. Units molt once; Lobster Knights molt twice.

- **Seafloor walking.** Land units cross shallows at full speed and can walk between islands joined by shallows without boats. Every map guarantees each home island a shallow path to at least one neighboring island, and the same path length for every player.
- **Brood Pools** (100 Brass, max 10) replace Farms: each makes 0.25 Food per second with no worker and no population.
- **Weaknesses.** Land units are 15% slower. No Torpedo Boats or Ironclads.

Unique unit	Era	Cost	Pop	HP	Attack	Range	Speed	Special
Spitter Shrimp	Sail	35 F, 30 B	1	45	6 pierce, reload 2.0	4	1.0	Hits cause Poison (1 dmg/s for 4 s)
Lobster Knight	Steam	70 F, 70 B	2	175	12 melee, reload 2.0	0.5	0.75	Armor 4/6; molts twice
Ghost Worm (sea)	Steam	140 F, 100 B	3	280	20 melee, reload 2.5	1	1.8	Swims; attacks ships and swimmers only; burrows (invisible) while idle; no Coal upkeep

- **Hero: The Lobster.** 700 HP instead of 500, the slowest hero. Active "Crushing Claw" (40 s cooldown): 80 damage to one target and stuns it 3 s.
- **Faction techs.** Chitin Plating (Steam, 200 F, 150 B): all units +1/+1 armor. Rapid Molting (Leviathan, 250 F, 200 B, 50 I): molt at 40 experience.
- **Ultimate: The Crustacean Queen.** Spawns 2 Spitter Shrimp every 10 s while summoned.
- **Wonder: The Great Brood Mound.** Units are trained already molted once.

Merfolk of the Weeds

Mermen, mermaids, and octopi. They own the sea between islands and strike from the kelp.

- **Swimming army.** Merfolk infantry and riders swim through any water at 1.2 speed, so they need no transports and have no landing penalty. On land they move 20% slower, and more than 6 tiles from the shore they deal 10% less damage.
- **Kelp Spread.** Kelp beds you own grow 1 tile every 15 s, up to 10 tiles out. On your kelp, Merfolk units are +20% faster and hidden unless an enemy is within 3 tiles; enemy ships on it are 20% slower.
- **Economy.** Kelp gathering +40%. Buildings can be placed on reefs and shallows.

Unique unit	Era	Cost	Pop	HP	Attack	Range	Speed	Special
Eel Rider	Steam	60 F, 50 B	2	115	9 melee, reload 1.8, +4 vs ranged, +8 vs ships	0.5	1.4 land, 2.2 water	Every 3rd hit stuns 1 s
Siren	Steam	40 F, 45 B	1	55	6 pierce, reload 2.0	6	1.0	Song: enemies within 5 tiles deal -20% damage (does not stack)

- **Hero: The Eel King.** Aura: swimming units +15% speed. Active "Riptide" (50 s cooldown): pulls enemy ships in a 5-tile circle toward the center and slows them 50% for 5 s.
- **Faction techs.** Octopus Ink (Steam, 150 F, 150 B): units leaving kelp stay hidden 5 s longer. Tidecallers (Leviathan, 250 F, 200 B, 50 I): Kelp Spread twice as fast and 14 tiles out.
- **Ultimate: Neptune.** A sea giant whose trident sends a wave that stuns every ship in a line.
- **Wonder: The Coral Throne.** Your kelp heals your units 2 HP/s.

Atlantean Lost Ones

The people trapped in the sunken city, worshipping something old. Weak for ten minutes, frightening after twenty.

- **Devotion.** A meter from 0 to 1,000. Gain +1 per second per relic held, +0.5 per second per ruin isle owned, and by offering units at a Chapel (half their cost in Devotion). Spend it on conversions and on the ultimate.
- **Awakening economy.** Gathering is 10% slower in the Sail Era, normal in Steam, and 10% faster in the Leviathan Era.
- **Weakness.** The slowest start in the game; rushes hurt.

Unique unit	Era	Cost	Pop	HP	Attack	Range	Speed	Special
Deep Priest	Steam	80 F, 50 B	1	40	—	7	0.9	Converts one enemy unit over 6 s for 30 Devotion; 20 s cooldown; cannot convert heroes, ultimates, or ships over 3 pop
Legionnaire (Giant Roman Armor)	Steam	70 F, 65 B	2	170	11 melee, reload 2.0	0.5	0.8	Armor 3/4; regenerates 1 HP/s

- **Hero: The Visitor** (the Alien). Aura: reveals hidden units within 8 tiles. Active "Phase Step" (60 s cooldown): teleports the hero and up to 6 nearby units 12 tiles.
- **Faction techs.** Whispers Below (Steam, 150 F, 150 B): Priests convert in 4 s. Drowned Legion (Leviathan, 250 F, 200 B, 50 I): converted units gain +20% HP.
- **Ultimate: Cthulhu Avatar.** Costs 150 Ichor and 600 Devotion instead of 300 Ichor. Its gaze Weakens every enemy on screen by 20% for as long as it stays.
- **Wonder: The Sunken Spire.** Devotion gain doubled.

Clockwork Foundry

The steampunk robots. An industrial army that never loses its nerve, as long as the furnaces burn.

- **Coal-powered army.** Military units replace their Food cost with Coal at 0.7× (a 40 Food, 30 Brass Rifleman costs 28 Coal, 30 Brass). Workers still cost Food.
- **Unbreakable.** Clockwork units ignore morale.
- **Salvage.** When any unit dies within 8 tiles of a Clockwork unit or building, the Foundry gains 12% of that unit's cost as Brass.
- **No Medics.** Mechanics (same stats as Medics) repair units instead.
- **Weakness.** If Coal hits 0, the whole army moves 40% slower and deals 25% less damage, not just ships. Storms deal double damage to Clockwork units.

Unique unit	Era	Cost	Pop	HP	Attack	Range	Speed	Special
Tesla Walker	Steam	100 B, 50 C	3	160	10 blast, reload 3.0	5	0.9	Lightning chains to 2 more targets at 40% damage; counts as siege
Rivet Golem	Steam	100 B, 80 C	3	300	16 melee, reload 2.2, +10 vs buildings	0.5	0.7	Armor 4/5

- **Hero: The Chief Engineer** (the Robot). Aura: nearby units repair 1 HP/s. Active "Overclock" (60 s cooldown): nearby units attack 30% faster for 10 s.
- **Faction techs.** Pressure Regulators (Steam, 200 B, 100 C): the Coal-shortage penalty is halved. Grounded Casings (Leviathan, 250 B, 150 C, 50 I): storms no longer hurt Clockwork units double, and electric attacks against them deal –25%.
- **Ultimate: The Steam Titan.** A walking fortress that fires cannons in all directions and carries 10 infantry.
- **Wonder: The Great Furnace.** Coal upkeep for all units –50%.

Neutral powers

Neutral sites are the map's third force: every one can be fought, and most can also be bargained with. Rewards are sized so that taking a site pays back its cost in about 4–6 minutes, and the riskiest sites pay the most.

Pirate Coves

Each cove is a fortress (4,000 HP) with 8 cutlass pirates, 4 shore cannons, and a harbor of pirate ships. Every player has a **reputation** with the pirates from –100 to +100, starting at 0.

Deal	Era	Price	What you get
Black market sell	Any	—	100 of any resource → 20 Doubloons
Black market buy	Any	60 Doubloons per 100 Food, Brass, or Coal; 100 Doubloons per 40 Ichor	The only way to buy Ichor

Deal	Era	Price	What you get
Cutter Raid	Sail	150 Doubloons	2 Sloops and 1 Gunboat attack your chosen target for 2 minutes
Pirate Squadron	Steam	350 Doubloons	4 Gunboats and a Transport that lands 10 pirates, for 3 minutes
Black Flag Armada	Leviathan	700 Doubloons	2 Ironclads, 4 Gunboats, and 16 pirates, for 3 minutes
Intel	Any	60 Doubloons (region) · 100 Doubloons (a player's base)	Reveals it for 30 s
Ransom	Any	Half the unit's cost in Doubloons	Buy back your units the pirates captured

- **Counter-bids.** The target of a raid gets a warning 20 s before the pirates arrive and can pay 125% of the contract to call them off, or 150% to send them back at whoever hired them. The hirer can top that bid by another 25%. Each raid contract can be bid on at most twice, so bidding wars end quickly.
- **Limits.** One raid contract per target at a time; each cove takes a new contract every 90 s. You cannot hire pirates against an ally.
- **Reputation.** Hiring +5, trading +1 per 100 resources, sinking pirate ships -30, attacking a cove -100. Every 25 reputation shifts prices by 5%. Below -50, that cove refuses you and its raiders target you when unhired.
- **Unhired pirates** raid the most valuable trade route on the map every 4 minutes (a Trade Ship route's owner gets a 20 s warning).
- **Storming the cove.** Destroy the fortress and defeat Blackbeard (1,500 HP; pistol volleys, calls 4 pirates every 30 s). The cove becomes yours: your deals cost 50% less, other players can still trade there at +25% and you receive that fee. It is a Pirate Ship nod from the platformer.
- **Ghost Pirate Ship** (minute 15, Medium and Large maps): a spectral ship roams the map, moving and attacking 2× faster than normal ships (1,200 HP). Sinking it pays 400 Doubloons and 1 relic.

Tribal Towns

Tribal towns guard fertile islands. A small town has a Longhouse (3,000 HP), 8 warriors, and 4 spear-throwers; a large town adds a mini-boss: the **Tribal Chieftain** (1,200 HP, war cry buffs warriors +20%) or the **Witch Doctor** (900 HP, poisons and heals tribesmen).

Option	Cost	Result
Conquer	An army of roughly 800–1,000 resources in Steam	Loot: 400 Brass, 200 Food, 80 Doubloons; the island is free to claim; 25% chance of a relic
Tribute	200 Food and 100 Brass every 5 minutes	Truce: the town stops raiding you and sells Tribal Warriors for 50 Food each (max 6 alive)
Kinship (Islanders only)	Shaman ritual, 150 Food and 100 Doubloons once	Full ally; see Islanders
Ignore	—	It raids you

- **Raids.** From minute 6, each town sends 4 warriors (Sail), 6 (Steam), or 8 plus a champion (Leviathan) every 5 minutes at the nearest player island it is not at peace with, or at whoever last attacked it.
- **Town growth.** Towns that are ignored add 1 warrior per minute (cap 16), so leaving them alone gets harder over time.

Volcano Islands

The highest-risk, highest-reward land on the map.

- **Rewards.** Two volcanic Ichor vents (3,000 Ichor each, gathered 60% faster than ruin vents) and the **summit altar**, which gives its owner +1 Ichor per second and is the objective of Volcano Ascension victory.
- **The Sun God** guards the altar: 4,000 HP, a fire-beam that deals 60 blast damage in a line, and a Strengthen aura on 6 fire imps. He reforms 8 minutes after defeat if nobody holds the altar.
- **Eruptions.** The volcano sleeps until minute 10, then erupts every 4 minutes. A 30 s tremor warning shows glowing cracks along 3 lava paths. Lava flows for 40 s at 40 damage per second and destroys buildings on its paths. An ash cloud follows for 60 s

(vision –50%). The vents always sit next to a lava path, so gathering there means pulling workers out on time.

- **Claiming the altar.** A unit must stand on it for 45 s uncontested after the Sun God is dead.

Other sites and events

Site or event	When	Risk	Reward
Shipwrecks	Always	20% are trapped: 3 drowned sailors or a burrowed Ghost Worm	600 Brass, 30 Doubloons
Sunken ruins	Always	4 Lost One sentinels (Steam-level)	1 relic, an 800-Ichor vent
Kraken	Wakes at minute 20 in the central sea	6,000 HP; grabs one ship at a time (30 dps) and sinks transports	The player who dealt the most damage gets 300 Ichor, 300 Doubloons, and Kraken Ink (their fleet is invisible for 60 s, once)
Storms, fog, whirlpools	Every 3 minutes (see Map)	Ship damage, blindness	Cover for raids
Pirate raids on trade	Every 4 minutes if unhired	Loses Trade Ships	—

All neutral strength scales with the lobby's Neutral Power setting: Low –30% HP and damage, High +30%.

Diplomacy, trade, and teams

In Open diplomacy each player sets a stance toward every other player; in Locked teams the stances are fixed at Allied between teammates and War with everyone else.

Stance	Units fight?	Can pass through territory?	Shared vision	Notes
War	Yes	Yes, by fighting	No	Default in free-for-all

Stance	Units fight?	Can pass through territory?	Shared vision	Notes
Ceasefire	No	No (units are pushed out)	No	Lasts 3 minutes, then returns to War unless changed
Peace	No	No	No	Allows trade routes and tribute
Open Harbors	No	Yes	No	Can dock and refuel (coaling range) at their Harbors
Allied	No	Yes	Yes	Shared vision, can garrison in each other's buildings, counts as a team for victory

- **Changing stance.** Offering a better stance needs the other player to accept. Downgrading is one-sided but announced to everyone, and breaking Allied or Peace into War gives the betrayer's units -15 morale and pauses their Trade Ships for 60 s, so a backstab is possible but costly.
- **Tribute.** Send any resource or Doubloons to a player at Peace or better; 10% is lost as a fee (0% for allies in Locked teams).
- **Trade routes.** A Trade Ship sails between your Exchange and another player's Exchange (Peace or better) or a pirate cove. Each round trip pays both ends Doubloons equal to $0.15 \times$ the distance in tiles (a 100-tile route pays 15 per trip; a round trip takes about 2 minutes, so 6 ships earn about 45 Doubloons per minute). Longer, more exposed routes pay more.
- **Communication.** Team and all chat, map pings (attack here, defend here, danger), and a diplomacy panel listing each player's stance, score, and islands, in the spirit of Call of War's diplomacy screen.
- **Team victory.** Allied players win together. In Open diplomacy, alliances must be at least 5 minutes old at the moment of victory to share it, so no one allies at the last second to claim a win.

Victory conditions and scoring

The first player or alliance to meet any enabled victory wins; if the time cap arrives first, the highest score wins. Every countdown below is shown to all players so everyone can react.

Victory	How to win	Countdown
Conquest	Destroy every rival's Harbors and Colony Halls. A player with no town center left has 90 s to land a Colony Ship and start a new one, or is eliminated	—
Relics	Hold every relic on the map (the count is shown in the top bar) garrisoned in your Chapels or Harbors	3 minutes
Volcano Ascension	Hold the summit altar through one full eruption cycle. On Large maps with two volcanoes, hold both altars	4 minutes
Score	Highest score when the time cap ends; ties go to most islands, then most relics	At the time cap

Wonders are not a victory on their own (to keep matches short); each is worth score and gives its faction passive.

Score

Source	Points
Each island you hold (Lighthouse, Harbor, or Colony Hall)	20
Each relic held	40
Enemy units killed	1 per 10 resources of their cost
Enemy buildings destroyed	1 per 20 resources of their cost
Resources gathered	1 per 100
Era reached	Steam 50, Leviathan 100
Technologies researched	5 each
Wonder standing	150
Neutral bosses	Blackbeard 50, Sun God 60, Kraken 80, tribal town conquered 25

Resigning. A player can resign at any time; their units and buildings are removed after 10 s. In team games, a team resigns only when every living member votes to.

Match pacing

A standard 4-player match should reach its deciding fights around minutes 20–25. Timers on neutral events are fixed so every match builds to the same climax.

1. **Minutes 0–3: opening.** Train workers, send the Scout Skiff out, fish, and build Cottages and a Farmstead or Mine Shed.
2. **Minutes 3–7: first expansion.** Claim the nearest islands with Lighthouses, especially the Coal isle. First pirate trades. Sail-era raids with Sloops, War Canoes, or Merfolk swimmers.
3. **Minute 6: tribes wake up.** First tribal raids. Decide to conquer, pay tribute, or ally them.
4. **Minutes 7–9: Steam Era.** Gunboats, Lancers, heroes, and Colony Ships arrive. Fights begin over ruin isles for the first Ichor.
5. **Minute 10: volcanoes wake.** First eruption. The Sun God becomes the mid-game prize.
6. **Minutes 10–16: the mid-game.** Naval battles over contested islands, Pirate Squadrons hired against the leader, alliances forming in free-for-all.
7. **Minute 15: Ghost Pirate Ship** appears on Medium and Large maps.
8. **Minutes 16–19: Leviathan Era.** Airships, elite units, and the first ultimates.
9. **Minute 20: the Kraken wakes** in the central sea.
10. **Minutes 20–30: endgame.** Relic and Volcano countdowns, final invasions of home islands, or the score finish at the time cap.

Faction timing windows (from the balance model): Islanders and Merfolk want to press in Sail and early Steam; Crustaceans and Clockwork are strongest from Steam onward; Nautilus and Atlanteans want the game to reach Leviathan.

Balance

Every counter in the shared roster wins an equal-cost fight with 57–66% of its army left, every unique unit sits within –27% to +10% of the shared units' cost-efficiency (lower where it has a strong ability), and the six factions land within $\pm 1.3\%$ of each other in overall strength. These numbers come from a combat simulator and a phase model built for this document; they are a starting point that playtesting must confirm.

How balance was checked

1. **Resource weights.** Food and Brass count as 1, Coal as 1.5 (fewer seams, and it is also upkeep), Ichor as 4 (few vents, needed to advance). A unit's cost value is the weighted sum of its cost.
2. **Cost-efficiency.** Efficiency = $\sqrt{\text{HP} \times \text{damage per second}} \div \text{cost value}$. Under Lanchester's square law this is what decides who wins a straight fight between equal budgets, so generalist units must sit close together.
3. **Equal-budget fights.** Two armies of the same resource value (1,200 land, 1,500 naval) fight in the simulator with armor, bonuses, range, closing speed, and melee crowding (only about 70% of a melee blob can reach the enemy at once). The target for a counter is 55–65% of its army surviving: a clear win, not a wipeout.
4. **Abilities modeled.** Chain lightning was modeled as $\times 1.8$ damage and the Siren's song as +25% effective HP; abilities that can't be simulated (invisibility, swimming, conversion) were paid for by lowering raw efficiency.
5. **Faction phase model.** Each mechanic was turned into an economy and army multiplier for each era, then weighted by era length (8, 10, and 12 minutes). Factions were tuned until overall strength matched within $\pm 3\%$ while their strongest era differs.

Shared unit results

Matchup (equal cost)	Winner	Winner's army left
Rifleman vs Harpooner	Rifleman	64%
Harpooner vs Lancer	Harpooner	57%
Lancer vs Rifleman	Lancer	60%
Bell Guard vs Rifleman	Bell Guard	66%
Harpooner vs Bell Guard	Harpooner	46%
Bell Guard vs Lancer	Bell Guard	43%
Lancer vs Mortar	Lancer	87% (siege is meant to die to riders)
Gunboat vs Torpedo Boat	Gunboat	65%
Ironclad vs Gunboat	Ironclad	63%
Torpedo Boat vs Ironclad	Torpedo Boat	65%

Every unit loses to at least one cheaper answer, and no shared unit beats every other one.

Unique unit results

Unique unit	Faction	Efficiency vs shared median	Paid for by	Beaten by (equal cost)
Nautilus Submarine	Nautilus	−27%	Invisibility while submerged	Gunboat (80% left), Torpedo Boat
Diving Marine	Nautilus	−9%	Swimming, no landing penalty	Lancer (60% left)
War Canoe	Islanders	+1%	Carries 5 infantry	Sloop (54%), Gunboat (70%)
Dog Handler	Islanders	+9%	— (cheap Sail unit, bleed)	Bell Guard (73%)
Spitter Shrimp	Crustacean	−15%	Poison	Lancer (70%)
Lobster Knight	Crustacean	+10%	Slow (0.75 speed)	Harpooner (36%)
Ghost Worm	Crustacean	+9%	Covers the missing Torpedo Boat and Ironclad	Torpedo Boat (70%)
Eel Rider	Merfolk	+3%	—	Harpooner (68%)
Siren	Merfolk	−20% (modeled)	Weakening song	Lancer (71%), Harpooner (33%)
Legionnaire	Atlantean	+7%	Slow, regenerates	Harpooner (52%)
Deep Priest	Atlantean	Not a fighter	Conversion costs Devotion	Lancers and anything fast

Unique unit	Faction	Efficiency vs shared median	Paid for by	Beaten by (equal cost)
Tesla Walker	Clockwork	-16% (modeled)	Chain lightning	Lancer (68%), Harpooner (25%)
Rivet Golem	Clockwork	+1%	Slow, costs Coal	Harpooner (45%)

Faction balance

Faction	Sail	Steam	Leviathan	Overall vs average
Nautilus Crew	0.96	1.04	1.15	-0.2%
Islanders	1.08	1.13	1.04	+1.3%
Crustacean Brood	0.98	1.07	1.09	-0.9%
Merfolk of the Weeds	1.09	1.12	1.02	+0.9%
Atlantean Lost Ones	0.90	1.05	1.19	+0.1%
Clockwork Foundry	0.92	1.09	1.10	-1.3%

Values are economy × army strength relative to a faction with no bonuses (1.00). Every faction has a weak era and a strong era, and the early aggressors (Islanders, Merfolk) are balanced against the late-game factions (Nautilus, Atlanteans) by the length of each era.

Other balance rules

- **Heroes and ultimates** have identical costs, HP, and damage budgets for every faction; ultimates with strong crowd control (Neptune's stun wave, Cthulhu's gaze) deal about 25% less damage to compensate.
- **Era timing.** An economy model puts a steady player in the Steam Era at about minute 9 and the Leviathan Era at about minute 18, matching the pacing targets.
- **Neutral rewards** pay back their cost in 4–6 minutes, so taking them is worthwhile but never automatic.

- **Six-player games** use a lower population cap (60) so that no single player can field an army large enough to end the game in one push, and so network traffic stays manageable.

Playtest tuning plan

1. Put every number in this document in one data file (for example `fathoms_balance.json`) that the game loads at startup, so tuning never requires recompiling.
2. Log every match: faction, winner, era timings, units built, and resources gathered.
3. After 20+ matches, any faction winning over 60% or under 40% of games gets one change of about 5% (a cost, HP, or gather rate), then retest.
4. Change one thing at a time and record it in a balance changelog.

Multiplayer and networking

Fathoms uses a host-authoritative model: one player's game (or a headless server) runs the real simulation, and everyone else sends commands and receives updates. LAN works through local discovery; friends far away connect through a small lobby and relay server, or through a virtual-LAN app while that server is being built.

How the pieces fit

- **Host (authoritative simulation).** Runs the full game at 20 ticks per second, validates every command (the player owns the unit, can afford it, the target is legal), and sends each client only what that client can see.
- **Clients.** Send commands (move, attack, build, train, research, diplomacy, chat). They render the world by smoothly interpolating between the host's updates about 100 ms behind, so movement looks smooth even with lag.
- **Lobby server.** A small program on a cheap cloud server. It creates join codes, tells players how to reach the host, helps connect through home routers (NAT hole punching), and relays traffic when a direct connection fails.
- **Headless server (optional, later).** The same game code built without graphics, running on the cloud server so no player has to host.

The arcade menu, lobby, chat, and connection code are one shared networking layer that all four Deep Arcade games use.

Connection modes

Mode	Who it's for	How it connects	Needs a server?
LAN	Players on the same network	Host broadcasts on the local network; joiners see it under Browse	No
Virtual LAN (testing)	Friends far away, early in development	Everyone installs ZeroTier or Tailscale, then uses LAN mode	No (free third-party app)
Direct IP	Friends far away, with a technical host	Host forwards one UDP port on their router; joiners type the host's address	No
Join code (main mode)	Friends far away	Lobby server matches the code to the host, punches through routers, relays if needed	Yes (small cloud server)
Steam (if ever released there)	Anyone	Steam's relay network and friends list	No (Steam provides it)

Technical requirements for Claude Code

- **Library.** Valve's open-source GameNetworkingSockets (C++): reliable and unreliable messages, encryption, and ICE/STUN support for connecting through routers. ENet is a simpler fallback.
- **Message types.** Reliable ordered: commands, chat, diplomacy, lobby changes. Unreliable: world updates (a lost one is replaced by the next).
- **World updates.** Delta-compressed: only units that changed since the client's last acknowledged update are sent. Positions quantized to 1/16 tile.
- **Fog-of-war filtering.** The host never sends a unit's position to a player who can't see it. This saves bandwidth and prevents map-hacking.
- **Bandwidth estimate.** A busy 6-player match (about 360 units, 150 buildings, neutrals) needs roughly 10–20 KB/s per client after filtering, so the host uploads about 1 Mbps. That fits normal home internet.

- **Responsiveness.** A click shows its marker and plays the unit's acknowledgment sound instantly on the client; the unit starts moving when the host confirms (typically 50–150 ms), which is normal for RTS games.
- **Version check.** Joining requires the same game build and the same balance-file hash, so no one plays with different numbers.
- **Disconnects.** The game pauses up to 2 minutes for a reconnect. After that, an AI takes over the player's faction (as Call of War does for inactive players) until they return.
- **Autosave.** The host saves the match every 3 minutes. If the host crashes, the host can reload the save and the same lobby code lets everyone rejoin.
- **Pausing.** Each player gets 3 pauses of up to 60 s per match.
- **Replays (later).** Recording the host's full world updates to a file allows watching a match again (a command-only replay would need a deterministic simulation, which this design avoids).

AI opponents

The AI fills empty lobby slots, replaces disconnected players, and lets you test the game alone. It plays by the same rules as humans (no extra resources or vision below Admiral) and issues the same commands a player would.

Difficulty	Reaction delay	Economy	Army behavior	Cheats
Deckhand	3 s	Reaches Steam around minute 12	Defends only; attacks after minute 20	None
Mate	1.5 s	Steam around minute 10	One attack wave per era; hires pirates rarely	None
Captain	0.8 s	Steam around minute 9	Scouts, counter-builds against what it sees, raids trade, uses heroes and ultimates	None
Admiral	0.4 s	Steam around minute 8	Multi-front attacks, bidding wars with pirates, times volcano eruptions	+15% gather rate

How it is built. A small set of managers that each run a few times per second:

1. **Economy manager.** Keeps workers training, balances gatherers against the targets in the Resources section, builds Cottages before hitting the pop cap.
2. **Expansion manager.** Scores nearby islands (resources, distance, danger) and sends Lighthouse builders.
3. **Military manager.** Tracks enemy unit types it has seen and builds the counter from the counter table.
4. **Neutral manager.** Decides whether to conquer, pay tribute to, or ally each tribe; trades with and hires pirates; avoids lava paths.
5. **Diplomacy manager** (Open diplomacy only). Offers peace to strong neighbors, allies against the score leader, and responds to human offers.
6. **Faction behaviors.** Each faction adds one rule set: Islanders prioritize Kinship, Merfolk fight on kelp, Crustaceans keep molted units alive, Atlanteans gather relics, Clockwork guards its Coal, Nautilus picks Logbook cards by what it's facing.

Build roadmap for Claude Code

Build Fathoms in eight stages, each ending in something playable, and only start a stage once the previous one's check passes.

1. **Foundations.** Balance data file, tile map, camera, selection, commands, and a fixed 20-tick simulation that takes commands as input (so networking plugs in later). *Check:* move workers around a hand-made island.
2. **Economy.** Gathering, drop-offs, Harbor, Cottages, farms, the Exchange, population. *Check:* reach the worker targets from the Resources section alone.
3. **Combat.** Shared land units, damage formula, counters, morale, the three eras, the Workshop techs. *Check:* equal-cost fights in-game match the Balance tables within about 10%.
4. **Islands and navy.** Archipelago generator with the fairness rules, ships, transports, landings, Lighthouses, coaling range. *Check:* a full 1v1 against a basic AI on a generated map.
5. **LAN multiplayer.** Shared arcade networking layer, lobby, host-authoritative updates, fog filtering, reconnects. *Check:* two computers on one network finish a match; then friends far away play it through ZeroTier or Tailscale.
6. **Factions, part 1.** Nautilus Crew, Islanders, and Clockwork Foundry, with heroes. *Check:* all three pairings played several times each.

7. **Neutral powers.** Tribes, pirates (including counter-bids), volcanoes, the Kraken, weather. *Check:* a 4-player match hits the Match pacing milestones.
8. **Factions, part 2, and internet play.** Crustacean Brood, Merfolk, Atlanteans, ultimates, wonders, diplomacy, the lobby/relay server with join codes, AI difficulty levels. *Check:* a 6-player match over the internet with join codes.

Tips for working with Claude Code on this

- Hand it this document and one stage at a time; ask it to write tests for the damage formula and the map fairness rules.
- Keep every number in the data file so balance changes never touch code.
- Ask it to port the balance simulator used for this document into a small command-line tool in the project, so any stat change can be checked against the Balance tables before a playtest.